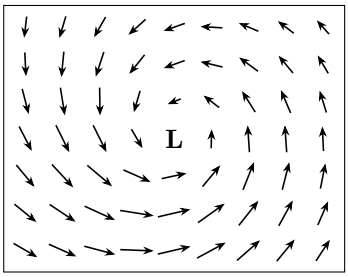
A New Kind of Function

So far this semester we have worked with two kinds of functions. A function like $f(x, y) = x^2 + y^2$ takes in a *point* and returns a *number*. A vector-valued function like $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ takes in a *number* and returns a *vector*.

Now think about a weather map that shows the wind. At every location there is a wind velocity, which has a speed and a direction, so it is a vector. The map assigns a vector to every point. The picture at the right is a simple model: air near a low-pressure center (L) circles counterclockwise and drifts inward, while the whole system moves east.



A model wind field around a low-pressure center

Scan for a live wind map centered on Flagstaff (earth.nullschool.net). The streaks follow the wind; faster streaks mean faster wind.

Water flowing in a river, the gravitational pull of the Earth, and the flow of heat through a metal plate all work the same way. A function that takes in a *point* and returns a *vector* is called a vector field.

Definition (Vector Field). A **vector field** in $\mathbb{R}^2$ is a function $\mathbf{F}$ that assigns to each point (x, y) in a set D a two-dimensional vector $\mathbf{F}(x, y)$. We write

$$\mathbf{F}(x, y) = \langle P(x, y),\, Q(x, y) \rangle = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}.$$

A **vector field** in $\mathbb{R}^3$ assigns to each point (x, y, z) in a set D a three-dimensional vector

$$\mathbf{F}(x, y, z) = \langle P(x, y, z),\, Q(x, y, z),\, R(x, y, z) \rangle = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}.$$

The set D is the **domain** of $\mathbf{F}$, and P, Q, R are its **component functions**.

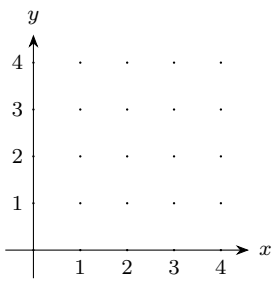
To draw a vector field, we draw the vector $\mathbf{F}(a, b)$ with its *tail at the point* (a, b), not at the origin. This is the key difference from how we drew position vectors at the start of the course.

Example 1. The velocity of water on the surface of a pond is modeled by $\mathbf{v}(x, y) = \langle y - x,\ xy - 1 \rangle$, measured in meters per second.

(a) Find the velocity vector at the points $(1, 2)$, $(3, 1)$, and $(0, 2)$.

(b) Draw each vector on the grid with its tail at its point.

(c) How fast is the water moving at $(3, 1)$?

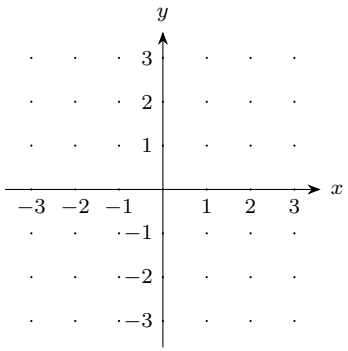


Sketching a Vector Field

A vector field has a two-dimensional input and a two-dimensional output, so its "graph" would need four dimensions. Instead, we draw the arrows right in the xy-plane. Choose a sample of points, compute the vector at each one, and draw it with its tail at the point. Use enough points to see the overall pattern, but not so many that the picture becomes a mess.

Example 2. Sketch the vector field $\mathbf{F}(x, y) = \langle -y,\, x \rangle$.

(x, y)	$\mathbf{F}(x, y)$	(x, y)	$\mathbf{F}(x, y)$
$(1, 0)$		$(-1, -1)$	
$(1, 1)$		$(0, -1)$	
$(0, 1)$		$(1, -1)$	
$(-1, 1)$		$(2, 0)$	
$(-1, 0)$		$(0, 2)$	



(b) Compute $\mathbf{F}(x, y) \cdot \langle x, y \rangle$. What does the answer tell you about the direction of each arrow?

(c) Compute $\|\mathbf{F}(x, y)\|$. Describe the vector field in words.

Two patterns show up again and again in this chapter.

Radial and Rotational Fields. In a **radial field**, every vector points directly toward or directly away from the origin, and the magnitude of a vector depends only on its distance from the origin. In a **rotational field**, the vector at (x, y) is tangent to the circle centered at the origin through (x, y), all vectors point clockwise or all point counterclockwise, and the magnitude of a vector depends only on its distance from the origin.

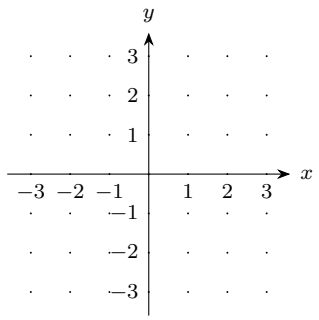
Sometimes only the *direction* of each arrow matters (for example, which way a fluid is flowing, not how fast). A vector field $\mathbf{F}$ is a **unit vector field** if $\|\mathbf{F}(x,y)\| = 1$ at every point. Dividing a field by its magnitude, $\mathbf{F}/\|\mathbf{F}\|$, is called **normalizing F**. It keeps every direction and makes every length 1. It only works at points where $\mathbf{F} \neq \mathbf{0}$.

Practice 1. Let $\mathbf{G}(x,y) = \langle x,\, y\rangle$.

(a) Sketch $\mathbf{G}$ by computing the vector at several points.

(b) Is $\mathbf{G}$ radial, rotational, or neither? Find $\|\mathbf{G}(x,y)\|$.

(c) Normalize $\mathbf{G}$. At which point is the unit vector field undefined?



Vector Fields in Space

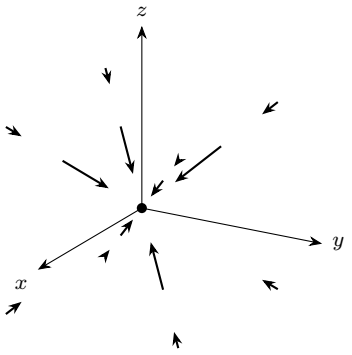
A vector field in $\mathbb{R}^3$ works the same way, with one more component function. Fields in space model gravity, electric and magnetic forces, and air or water flowing through a region. They are hard to sketch by hand, so we usually describe them in words and let a computer draw them.

Example 3. A mass M sits at the origin and a mass m sits at the point (x,y,z). Let $\mathbf{r} = \langle x,y,z\rangle$ be the position vector of the second mass. Newton's law of gravitation says that the force the first mass exerts on the second is $\mathbf{F}(x,y,z) = -\dfrac{GMm}{\|\mathbf{r}\|^3}\,\mathbf{r}$, where G is the gravitational constant.

(a) In which direction does $\mathbf{F}(x,y,z)$ point?

(b) Show that $\|\mathbf{F}\| = \dfrac{GMm}{\|\mathbf{r}\|^2}$.

(c) Is $\mathbf{F}$ a radial field? What is its domain?



Gradient Fields

We have actually met vector fields before. When we studied directional derivatives, the gradient of $f(x, y)$ was

$$\nabla f(x, y) = f_x(x, y)\,\mathbf{i} + f_y(x, y)\,\mathbf{j}.$$

The gradient assigns a vector to every point, so ∇f is a vector field. Recall two facts: ∇f points in the direction in which f increases fastest, and ∇f is perpendicular to the level curves of f. The same is true in three dimensions, where $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j} + f_z\,\mathbf{k}$.

Definition (Conservative Field and Potential Function). A vector field $\mathbf{F}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is a **gradient field**, or a **conservative vector field**, if there is a scalar function f with

$$\mathbf{F} = \nabla f.$$

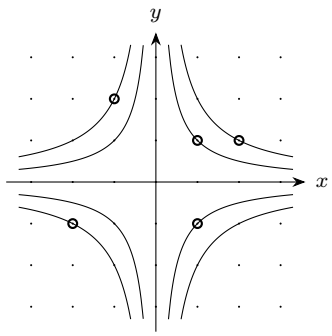
The function f is called a **potential function** for $\mathbf{F}$.

The name "conservative" comes from physics: these fields model forces, like gravity, in which energy is conserved. Conservative fields will be the stars of the rest of this chapter.

Example 4. Let $f(x, y) = xy$.

(a) Find the gradient field ∇f.

(b) The figure shows the level curves $xy = \pm 1$ and $xy = \pm 2$. Draw ∇f at the marked points $(1, 1)$, $(2, 1)$, $(-1, 2)$, $(1, -1)$, and $(-2, -1)$. What do you notice?



Because ∇f is built from derivatives, finding a potential function is a little like finding an antiderivative, and the same thing happens with constants: if f is a potential function for $\mathbf{F}$, then so is $f + C$ for any constant C. To *check* whether f is a potential function for $\mathbf{F}$, compute ∇f and compare it to $\mathbf{F}$, one component at a time.

Example 5. Is $f(x, y, z) = x^2y + yz^3$ a potential function for $\mathbf{F}(x, y, z) = \langle 2xy,\ x^2 + z^3,\ 3yz^2 \rangle$?

Practice 2.

(a) Is $f(x, y) = x^2 \cos y$ a potential function for $\mathbf{F}(x, y) = \langle 2x \cos y,\ x^2 \sin y \rangle$?

(b) Find the gradient field of $f(x, y, z) = xe^{yz}$.

A Test for Fields That Are Not Conservative

Most vector fields are *not* conservative. Here is a quick way to catch many of them. Suppose $\mathbf{F} = \langle P, Q \rangle$ is conservative, so $\mathbf{F} = \nabla f$. Then $P = f_x$ and $Q = f_y$, so

$$P_y = f_{xy} \qquad \text{and} \qquad Q_x = f_{yx}.$$

Clairaut's Theorem says the mixed partials f_{xy} and f_{yx} are equal, so $P_y = Q_x$.

The Cross-Partial Property of Conservative Fields. Suppose the component functions of $\mathbf{F}$ have continuous first partial derivatives.

- If $\mathbf{F} = \langle P, Q \rangle$ is conservative, then $P_y = Q_x$.
- If $\mathbf{F} = \langle P, Q, R \rangle$ is conservative, then $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$.

So if any cross-partial pair is *not* equal, then $\mathbf{F}$ is *not* conservative.

A warning. The test only works in one direction. If the cross-partials are all equal, the test by itself does *not* tell us that $\mathbf{F}$ is conservative. We will learn how to finish that job later in this chapter.

Example 6.

(a) Show that the rotational field $\mathbf{F}(x, y) = \langle -y,\ x \rangle$ from Example 2 is not conservative.

(b) Is $\mathbf{F}(x, y, z) = \langle 2xy,\ x^2 + z,\ yz \rangle$ conservative?

Practice 3.

(a) Show that $\mathbf{F}(x,y) = \langle e^y,\ xe^y + x\rangle$ is not conservative.

(b) Let $\mathbf{F}(x,y) = \big\langle x + y^2,\ 2xy + 1\big\rangle$. Do the cross-partials show that $\mathbf{F}$ is not conservative? Then check whether $f(x,y) = \frac{1}{2}x^2 + xy^2 + y$ is a potential function for $\mathbf{F}$. What can you conclude?

Answers

Example 1. (a) $\mathbf{v}(1,2) = \langle 1,1\rangle$, $\mathbf{v}(3,1) = \langle -2,2\rangle$, $\mathbf{v}(0,2) = \langle 2,-1\rangle$. (c) $\|\langle -2,2\rangle\| = 2\sqrt{2} \approx 2.83$ m/sec.

Example 2. (a) $\langle 0,1\rangle$, $\langle -1,1\rangle$, $\langle -1,0\rangle$, $\langle -1,-1\rangle$, $\langle 0,-1\rangle$ (left column); $\langle 1,-1\rangle$, $\langle 1,0\rangle$, $\langle 1,1\rangle$, $\langle 0,2\rangle$, $\langle -2,0\rangle$ (right column). (b) $\mathbf{F}\cdot\langle x,y\rangle = -xy + xy = 0$, so each arrow is perpendicular to the line from the origin, which means it is tangent to the circle through its point. (c) $\|\mathbf{F}\| = \sqrt{x^2+y^2}$, the distance to the origin. The arrows circle the origin counterclockwise and get longer farther out.

Practice 1. (b) Radial, pointing away from the origin; $\|\mathbf{G}\| = \sqrt{x^2+y^2}$. (c) $\dfrac{\mathbf{G}}{\|\mathbf{G}\|} = \dfrac{1}{\sqrt{x^2+y^2}}\langle x,y\rangle$, undefined at the origin.

Example 3. (a) $\mathbf{F}$ is a negative multiple of $\mathbf{r}$, so it points from (x,y,z) toward the origin. (b) $\|\mathbf{F}\| = \big(GMm/\|\mathbf{r}\|^3\big)\|\mathbf{r}\| = GMm/\|\mathbf{r}\|^2$. (c) Yes: it points toward the origin and its magnitude depends only on the distance $\|\mathbf{r}\|$. Domain: all of $\mathbb{R}^3$ except the origin.

Example 4. (a) $\nabla f = \langle y,x\rangle$. (b) $\langle 1,1\rangle$ at $(1,1)$, $\langle 1,2\rangle$ at $(2,1)$, $\langle 2,-1\rangle$ at $(-1,2)$, $\langle -1,1\rangle$ at $(1,-1)$, $\langle -1,-2\rangle$ at $(-2,-1)$. Each arrow is perpendicular to the level curve through its point.

Example 5. Yes: $\nabla f = \big\langle 2xy,\ x^2+z^3,\ 3yz^2\big\rangle = \mathbf{F}$.

Practice 2. (a) No: $\nabla f = \big\langle 2x\cos y,\ -x^2\sin y\big\rangle$; the second component has the wrong sign. (b) $\nabla f = \langle e^{yz},\ xze^{yz},\ xye^{yz}\rangle$.

Example 6. (a) $P_y = -1$ and $Q_x = 1$ are not equal, so $\mathbf{F}$ is not conservative. (b) $P_y = 2x = Q_x$ and $P_z = 0 = R_x$, but $Q_z = 1$ while $R_y = z$. Not conservative.

Practice 3. (a) $P_y = e^y$ and $Q_x = e^y + 1$ are not equal, so not conservative. (b) $P_y = 2y = Q_x$, so the test does not rule it out (and does not prove anything by itself). But $\nabla f = \big\langle x+y^2,\ 2xy+1\big\rangle = \mathbf{F}$, so $\mathbf{F}$ is conservative.